Final Project: Trajectory Analysis Presentation and Paper

This final project is designed to directly assess your mastery of our core learning outcomes: your ability to process unstructured spatial data, and extract meaningful behavioral phenotypes or system demands. You must identify a specific, solvable spatial problem, build a mathematically rigorous pipeline to address it, and empirically defend your engineering choices.

Objective: Synthesize the concepts of movement modeling, spatial data preprocessing, and algorithm evaluation into a comprehensive research project. You will identify a specific problem in the domain, build a data pipeline to address it, evaluate your results, and present your findings in both a live presentation and a formal academic paper.

Part 1: Presentation

The presentation serves as the preliminary defense of your pipeline and methodology. It is an opportunity to communicate your core problem, demonstrate your data processing strategy, and receive peer and instructor feedback before finalizing your written paper.

Presentation Specification (100 Marks)

Category	Description	Weight
Problem & Solution	Clear articulation of the specific problem being solved. The proposed algorithmic or modeling solution is appropriate and mathematically sound for the movement data context.	20%
Data Pipeline	Detailed explanation of the data pipeline. Must include the identification and handling of spatial outliers, and the appropriateness of the chosen filtering, temporal stratification, and aggregation preprocessing steps.	20%
Experimental Setup	Completeness of the experimental setup. Variables, algorithms, and baseline comparisons are clearly defined so that the experiment could theoretically be replicated.	20%
Results	Quality of the results presentation. Graphs clearly demonstrate the performance of the algorithms or heuristics under the stated conditions.	20%
Delivery & Q&A	Quality of the presentation (clarity, pacing, slide design) and the ability to effectively and directly answer questions from the audience.	20%

Part 2: The Academic Paper

The final paper must adhere to the structural conventions of a computer science academic, geophysical or data science paper. Your paper should include the sections covered in class. Evaluation will focus on the articulation of the specific problem statement, the rigour of presentation of the data pipeline, the appropriateness of model chosen, and the quality of the results and their presentation. You should be able to demonstrate that you are able to take a question about mobility, frame it as a data pipeline, clean the data for analysis, document the results of that analysis, and draw appropriate conclusions about reality from the analysis.

Paper Rubric (100 Marks)

Category	Description	Weight
Introduction	By the end of the introduction, the reader should understand what you did, why you did it, and what the significance is. The question and the appropriateness of the solution are clearly established.	20%
Setup & Data Pipeline	Describes the experimental setup as succinctly and precisely as possible. Fully explains the data pipeline, including outlier identification, filtering, stratification, and aggregation preprocessing.	15%
Results Presentation	Results are the proof part of the paper. Each graph must be introduced with what it is supposed to demonstrate. Each graph must be closed with how it demonstrates the point. Any interesting or anomalous behavior must be noted.	25%
Other Sections	Evaluates the Discussion, Abstract, Literature Review, and Conclusion. The Discussion section must include 2-3 paragraphs explaining why failures occurred in your pipeline, what you would do to fix them, and how you would address them in future work.	20%
Writing & Format	Written quality, academic tone, and adherence to the specified section formats.	10%
Feedback Integration	Demonstrates that you actively addressed and integrated any critiques and issues identified during the live presentation Q&A.	10%